

ANKITH REDDY SUBHANPURAM

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LinkedIn | GitHub | Portfolio

PROFESSIONAL EXPERIENCE

University of Florida – Intelligent Clinical Care Center (PRISMAp)

Research Assistant

Gainesville, FL

Aug 2025 – Present

- Engineered synthetic clinical (EHR) datasets using frontier models from clinically validated data for hospital deployment.
- Implemented an embedding-based deduplication pipeline leveraging FAISS and Jaccard similarity, reducing dataset size by 36% and improving model accuracy by 4%.
- Fine-tuned open-source LLMs using Axolotl on HiPerGator's 4 NVIDIA B200 GPUs with FSDP2 for efficient training.
- Developed an LLM-as-judge evaluation framework with 14 judges validating 110K EHR samples for clinical reasoning and diagnostic accuracy.
- Orchestrating end-to-end MLOps pipelines using PyTorch, HuggingFace, MLflow, and Docker on NVIDIA DGX systems.

Scholarship Auditions

Artificial Intelligence Engineer Intern

Nashville, TN

May 2025 – Aug 2025

- Architected a Retrieval-Augmented Generation (RAG) chatbot using the Azure AI stack, Azure Cosmos DB (NoSQL) as a vector store, and OpenAI APIs for context-aware responses across 20K+ documents.
- Implemented hybrid retrieval with embedding-based search and reranking models, improving retrieval relevance by 25% and reducing response latency by 35%.
- Fine-tuned ResNet-50 model for music transcription image classification, achieving 90%+ accuracy on 10K+ annotated images through transfer learning and domain-specific optimization.
- Developed automated web scraping pipelines using Scrapy and Selenium to collect and structure data from 50+ sources, generating over 100K records.

Centre for Artificial Intelligence and Robotics (DRDO)

Artificial Intelligence Research Intern

Bengaluru, India

Oct 2023 – Mar 2024

- Engineered Automatic Speech Recognition (ASR) models based on Wav2Vec2, leveraging DGX GPUs and distributed data parallel training in PyTorch, reducing training time by 40% on 1K+ hours of speech data.
- Implemented Byte Pair Encoding (BPE) for multilingual vocabulary optimization, reducing vocabulary size by 30% while improving tokenization efficiency.
- Improved recognition performance for low-resource languages (Assamese, Bengali) through fine-tuning and hyperparameter optimization, achieving 8.3% Word Error Rate (WER).
- Developed Voice Activity Detection (VAD) and noise reduction pipelines, improving ASR robustness and increasing transcription accuracy by 15% in noisy environments.

SKILLS

Programming: Python, C, SQL

Professional Skills: Machine Learning, Artificial Intelligence, Data Science, Data Analytics, NLP, LLMs, RAG Systems, LLM Fine-tuning, GenAI, MLflow, Docker, Git, Apache Airflow, NVIDIA DGX, Distributed Data Parallel (DDP), Selenium, Scrapy, Web Scraping, Deep Learning, Speech Recognition, Audio Processing, Computer Vision

Frameworks: PyTorch, Scikit-Learn, TensorFlow, Pandas, NumPy, NLTK, SpaCy, Keras, Librosa, HuggingFace

EDUCATION

University of Florida

Master of Science in Artificial Intelligence Systems
Cumulative GPA: 3.89 / 4.00

Gainesville, FL
Aug 2024 – May 2026

Vignana Bharathi Institute of Technology

Bachelor of Technology, Computer Science and Engineering (AI & ML)
Cumulative GPA: 8.76 / 10.0

Telangana, India
May 2020 – May 2024

PROJECTS

MSAIS Program Assistant

Designed a conversational AI assistant implementing a RAG pipeline using LangChain, FAISS, and HuggingFace. Scraped and structured data using Scrapy, deployed on HuggingFace Spaces with a Gradio interface, powered by Llama-3.3-70B.

Personal AI Assistant

Built an autonomous personal assistant using LangGraph agents and OpenAI capable of executing tasks from natural language prompts including email automation, calendar creation, and social media publishing.

Virtual Surya Namaskar Coach

Designed a movement tracking system using OpenCV to analyze yoga postures and identify common mistakes during online sessions. Technologies used: Python, OpenCV, TensorFlow.

LEADERSHIP & VOLUNTEER EXPERIENCE

EpsilonPI, ML Club in VBIT

ML Associate

Hyderabad, India
Jul 2023 – May 2024

- Led a team of four members in research and AI application development for university initiatives.